# Cloning-free long-read profiling of somatic hypermutation

Two biological systems test one control-based Oxford Nanopore workflow

## 1 Complementary test systems

### Mouse GC B cells: $J_{H4}$ downstream intron

~1.23-kb amplicon

Wt ($n = 3$)

*Aicda*$^{-/-}$ ($n = 3$)

Same mice benchmarked by cloned-amplicon Sanger and ONT

### N114 hybridoma

**Fixed immunoglobulin V region**

N114

pMX-*Aicda*

pMX-Empty vector

~0.65-kb amplicon

$n = 3$ independent subclones per condition

## 2 Shared ONT workflow

**R10.4.1 | Kit 14 | SUP basecalling**

Intact amplicons are retained as single-molecule reads



substitutions remain phased within each read

**Read QC**
length ≤ max
EE ≤ 0.01/base

**Align**
vs. paralogue;
chimeras
removed

**Depth**
random
500 reads

**Call SNVs**
tested against
control errors

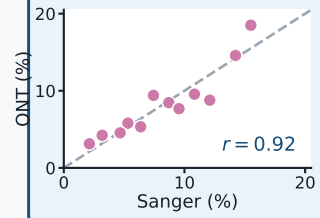
**SHMtrace: Benjamini–Hochberg FDR ≤ 0.05; sites at ≥ 50% removed**
**Each position–substitution counted once per sample**

### Control-based interpretation

*Aicda*$^{-/-}$ or empty-vector controls
measure residual background

## 3 Validated outcomes

### Sanger concordance




### AID dependence

**Mouse $J_{H4}$** **≥141×**

**N114 V region** **≥85×**

G:C-biased AID signature

### Longer region

$J_{H4}$: 1.23-kb intact reads

529-bp Sanger window

**1,141-bp reliable analysis**

long-range linkage retained

### Depth + background

**93%** $J_{H4}$ **74%** N114

variants recovered at 500 reads

*Aicda*$^{-/-}$ control background
**≈0.48 × 10$^{-5}$ variants/bp**

**Cloning-free • intact long molecules • depth-standardised, substitution-focused profiling**